

Scabo: XR Smart Cardboard Toy Kit

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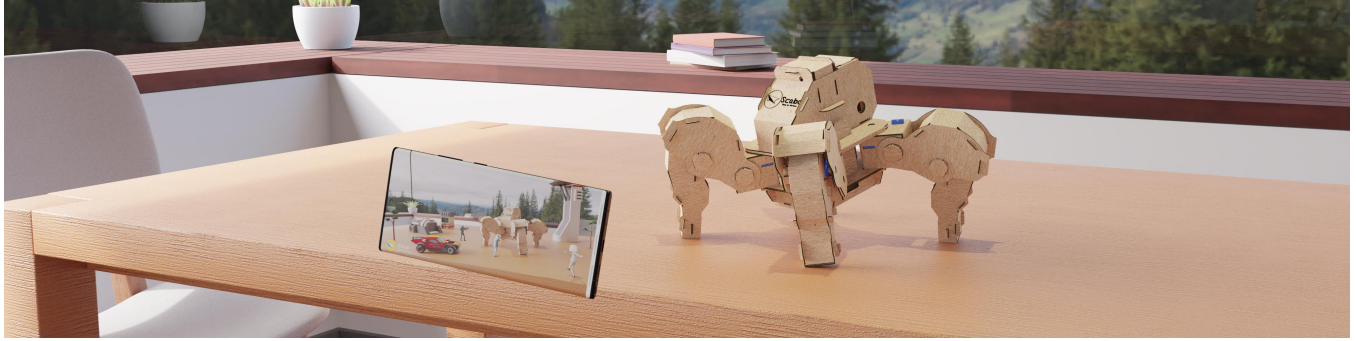


Figure 1: Playing AR game with the Scabo robot.

ABSTRACT

We provide a new eco-friendly solution to the toy industry, which includes both hardware and software integration: a cardboard robot kit and an AR platform. This solution offers users a D-I-Y experience of making a legged robot from scratch as well as interacting with the AR scenarios. We utilize both AI and XR technologies to digitalize the real environment and then enhance users' experiences.

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1 INTRODUCTION

The toy industry is the most plastic intensive industry in the world, according to a UN environment programme report in 2014. In addition, the number of toys is countless, but most of them have short lifetimes due to limited contents. In the Labs Installation session this year, we introduce a new toy kit: *Scabo robot kit*, made of 100 % recycled material, cardboard. Users can have a D-I-Y experience

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building their own legged robot from scratch and engage in immersive XR scenarios.

2 DESIGN AND IMPLEMENTATION

D-I-Y experience. A Scabo robot kit comes as a collection of cardboard, electronic modules and jumper cables (Figure 2). The cardboard papers are pre-cut to be detached easily by hand without the need for a scissor. Each part is designed to be constructed and assembled to the final robot without requiring any glue or screw. Assembly instruction manuals are depicted as animated 3D models.

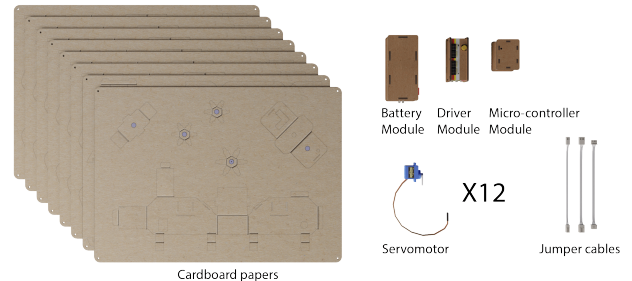


Figure 2: Parts inside a Scabo robot kit.

Hardware specific. The electronic parts include a battery module, a driver module, a micro-controller module and 12 servomotors.

The battery module holds a li-ion battery, which provides 8.4V for the driver module to control 12 servos of 4 legs. The battery box also supplies a voltage of 5V for an MCU node inside the micro-controller box, which takes responsibility to communicate with Android/ iOS devices through Bluetooth Low Energy (BLE) and Wi-Fi. Command signals from mobile devices and the robot's status are all transmitted through these connections.

XR experience. We utilize the power of Google's platform ARCore¹ and Sceneform² to build augmented reality experiences in Android devices. In terms of iOS devices, ARKit³ and SceneKit⁴ are employed. Experiencing XR includes three steps. First, attendees use smart devices supporting AR to detect flat surfaces like a table or a floor. Then, the robot's initial location on the flat surfaces is estimated by our robot tracking algorithm. Lastly, 3D scenarios are rendered on the smart devices' screens. To enhance players' experiences, computer vision, collision detection and response algorithms are used to support the interaction between virtual characters and physical cardboard robots in XR environments. Users can either entertain by playing AR games or learn science subjects through edutainment games with the robot.

Technology. To trigger virtual characters' and robots' behaviours when they contact, their bounding boxes in a 3D environment are required beforehand. Regarding virtual characters in specific and other 3D models in general, the bounding boxes are available, in contrast to that of the robot which is not known. The most common solution for this is using predefined formal markers, but the algorithm occasionally fails due to bad lighting conditions, motion blurs, maker deformations or occlusions. Applying 3D detection deep learning models [Ahmadyan et al. 2020] can solve this drawback and reach higher accuracy, however, require powerful hardware which is not appropriate for the majority of commercial phones in a mid-end segment having limited computing resources. To balance accuracy and processing time, we employ a lightweight 2D CNN detection model: MobileNetV2 [Sandler et al. 2018] to detect the robot, and then apply a mapping algorithm from 2D to 3D by casting a ray from the camera viewpoint through the bounding box center to the detected surface. This combination runs seamlessly in these smart devices. In addition, natural processing language modules will take responsibility for conversational interaction between users and the robot.

Demo: D-I-Y and AR Games. In the demo, attendees will participate in a D-I-Y activity, which involves building a cardboard robot from scratch. After the assembly, the robot can then be played as a part of our team shooting game, Cardboard Blitz War (Figure 3). The main objective is to control the robot, fight against enemies, defeat them all and survive within the allotted time (3 minutes). In addition, attendees can also explore scientific AR scenarios, and talk to virtual scientists to learn more about space, geography, anatomy, etc (Figure 4).



Figure 3: Cardboard Blitz War

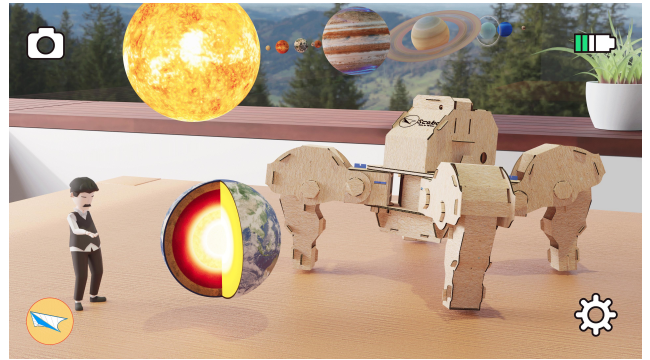


Figure 4: Cardboard Edutainment

3 CONCLUSION

We have introduced a new eco, user-friendly toy kit, Scabo Robot. Our solution offers users a D-I-Y experience of assembling a fully functional cardboard robot from scratch, as well as an immersive XR gaming and learning experience which comes along with the toy kit.

Currently, our application only supports the interaction between virtual characters and the cardboard toy kit in AR scenarios, but not with already available toys in the house. In the future, we intend to further develop this feature which enables these toys to be a part of the virtual environments. This allows children to play with any toy they have in the AR environment.

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¹ARCore: <https://developers.google.com/ar>

²Sceneform: <https://developers.google.com/sceneform>

³ARKit: <https://developer.apple.com/documentation/arkit>

⁴SceneKit: <https://developer.apple.com/documentation/scenekit>